Rework Plan — Spike Not Feasible

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What to do when the spike fails its go / no-go rules: send the project back to step 1 with changed requirements, or stop it.

When this applies

Rework is triggered when a core experiment fails and no fallback in the spike's go / no-go rules covers it.

Failed experiment	Rework?	Why
E1 End-to-end offline run	Yes	The product's core promise (offline dubbing) is not proven
E5 Diarization below 90%	Yes	Multi-speaker voices (BR-05) are unreliable
E7 Sync: fewer than 95% of lines within 0.5 s	Yes	Dub feels wrong to viewers
E2 Translation below 3.5/5, and the local LLM also fails	Yes	Viewers cannot follow the story
E2 fails but the local LLM passes	No	Go with the LLM; raise hardware needs
E4 Voice not Bangladeshi enough	No	Go with presets; fine-tune later
E6 Too slow on CPU only	No	CPU stays pre-process only
E3 or E8 fails alone	Product owner decides	Narrows scope rather than blocking it

Rework process

One rework loop should take no more than 5 working days before a new spike starts.

1. **Record the failure.** Fill the rework record (template below) with the experiment, measured result, test clip and machine.

2. **Find the cause.** Use the failure playbook to separate a tuning problem (fixable in the spike) from a limit of today's models.
3. **Retry once if it is tuning.** A tuning fix (a setting, a different model size, a hint like `--speakers`) gets one re-run inside the spike timebox. If it passes, continue to Development without rework.
4. **Choose a rework option.** For a model limit, pick an option from the playbook with the product owner.
5. **Update the BRD (step 1).** Change the affected requirement, priority or target, and give the BRD a new version and date.
6. **Update the gap analysis (step 2).** Re-score feasibility for the changed requirements only.
7. **Re-run the spike (step 3)** for the failed experiments only, plus E1 end to end.
8. **Limit the loops.** After 2 rework loops on the same experiment, the product owner decides between further rework and "not feasible".

Failure playbook

Failure	Likely cause	Tuning fix (retry once)	Rework option	BRD change
E1 offline run fails	A model still calls the network; missing dependency; out of memory	Set offline flags; pre-download all models; use smaller models	Replace the failing component with one that runs offline	None if replaced; otherwise BR-07 allows a one-time online setup step
E5 diarization below 90%	Overlapping speech, many speakers, noisy audio	Pass the speaker count; run on separated vocals	Let users merge or relabel speakers in a review step (BR-15)	BR-05: limit to 4 speakers; add a manual speaker review step
E7 sync below 95%	Bangla lines longer than the original	Raise speed-up cap to 1.5×; reduce TTS pauses	Ask the LLM for shorter Bangla that fits the time slot	NFR-05: relax to 1.0 s; allow slight overlap into silence

Failure	Likely cause	Tuning fix (retry once)	Rework option	BRD change
E2 translation below 3.5/5	Literal machine translation; English pivot loses meaning	Use larger IndicTrans2 (1B); fix sentence splitting	Local LLM translation or post-editing; Bangladeshi glossary	NFR-02: raise minimum hardware for the LLM; BR-15 to Should
E3 non-English pivot poor	Whisper translate-to-English loses nuance	Use Whisper large-v3, not turbo	Direct Hindi → Bangla with IndicTrans2 Indic-Indic; LLM for others	Narrow launch languages to English and Hindi
E8 background not clean	Demucs artefacts on dense music	Try another Demucs model	Duck the original audio instead of removing voices	BR-06 to Should
E6 too slow on recommended GPU	Model sizes too large	Whisper turbo; int8; batch TTS	Pre-process only, drop live mode from version 1	BR-09 moves to version 2

Not feasible: when to stop

The product owner declares the project not feasible if any of these hold after the rework loops:

- E1 still fails: no offline chain works on the recommended hardware.
- E2 and E5 both fail after 2 loops: the dub would be neither understandable nor correct per speaker.
- The only working option breaks a constraint that cannot change: it needs internet, a non-commercial model for a commercial product, or hardware the target users do not own.

When stopping, record:

- Final results for every experiment and the hardware used.

- Which component blocked it, and what would need to change (a new model release, a Bangladeshi dataset, cheaper GPUs).
- A review date to re-check the blocking component, for example 6 months later.
- Any reusable parts, such as the player and mixing code, which work on their own.

A partial release is an alternative to stopping: for example, Bangla subtitles only (BR-13) or single-speaker dubbing, if the product owner accepts the reduced scope.

Rework record template

Copy one table per rework loop.

Field	Entry
Rework loop #	
Date	
Failed experiment(s)	
Measured result vs pass condition	
Test clip and machine	
Cause (tuning or model limit)	
Tuning retry result	
Rework option chosen	
BRD changes (requirement ID, old → new)	
BRD new version	
Approved by (product owner)	
Re-spike result	
Decision	Go / rework again / not feasible